

Probability PhD Exam, May 2005

1. Let X_1, X_2 be two independent random variables with the same uniform distribution on $(\theta - 1/2, \theta + 1/2)$, and let $Y_1 = \min(X_1, X_2)$, $Y_2 = \max(X_1, X_2)$,
 - (a). Find $\mathbb{P}(Y_1 \leq \theta \leq Y_2)$,
 - (b). Find $\mathbb{P}(Y_1 \leq \theta \leq Y_2 \mid Y_2 - Y_1 \geq 1/2)$.

2. Let $\phi(t)$ be a characteristic function, prove that

- (a). $1 - \operatorname{Re}(\phi(2t)) \leq 4(1 - \operatorname{Re}(\phi(t)))$,
- (b). $1 - |\phi(2t)|^2 \leq 4(1 - |\phi(t)|^2)$.

3. Let $\{X_n, n \geq 1\}$ be a sequence of i.i.d. nonnegative random variables. let $S_0 = 0$, and $S_n = X_1 + \dots + X_n$. For $t > 0$, we define

$$\{\omega \mid N_t(\omega) = n\} = \{\omega \mid S_n(\omega) \leq t < S_{n+1}(\omega)\}$$

show that

$$\mathbb{E}(N_t(\omega)) = \sum_{n=1}^{\infty} \mathbb{P}(S_n(\omega) \leq t).$$

4. Let $X_1, X_2, \dots$ be a sequence of strictly positive random variables such that

$$\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = f_n(X_n).$$

For $n \geq 2$, let

$$M_n = \frac{X_1 X_2 \cdots X_n}{f_1(X_1) f_2(X_2) \cdots f_{n-1}(X_{n-1})}.$$

- (a). Show that for $n \geq 2$, M_n is a $\mathcal{F}_n$ -martingale.
 - (b). Does this martingale converges almost surely and in L^1 ? Explain it.
5. Let $\{M_n, n \geq 0\}$ be a sequence of integrable random variables adapted to a filtration $\mathcal{F}_n$. Assume that for each bounded stopping time T , $\mathbb{E}(M_T) = \mathbb{E}(M_0)$, show that $\{M_n, n \geq 0\}$ is a martingale.
 6. Let X and Y be random variables such that $\mathbb{E}(X^2) < \infty$ and $\mathbb{E}(Y^2) < \infty$, $\mathbb{E}(X \mid Y) = Y$ and $\mathbb{E}(Y \mid X) = X$. Show that $X = Y$ a.s.
 7. Let X, Y be two independent random variables with $\mathbb{E}[Y] = 0$. Show that for $p \geq 1$

$$\mathbb{E}[|X|^p] \leq \mathbb{E}[|X + Y|^p].$$

8. Let (X, Y) be a random point on a unit circle with uniform distribution, that is

$$\mathbb{P}((X, Y) \in A) = \frac{\text{length}(A)}{2\pi}$$

for any Borel subset A of $C_2 = \{(x, y) \mid x^2 + y^2 = 1\}$. Find the marginal distribution of X .

9. Let $\mathcal{F}_n$ be a filtration, $|X_n| \leq Y$, Y integrable. Suppose that $X_n \rightarrow X$ a.e. Using the martingale convergence theorem to prove that

$$\mathbb{E}[X_n \mid \mathcal{F}_n] \rightarrow \mathbb{E}[X \mid \mathcal{F}_\infty] \quad \text{a.e.}$$

10. Let $Y \in L^p$, $|X_n| \leq Y$ and $X_n \rightarrow X$ in distribution. Show that X_n converges to X in L^p .